A Machine-Checked Reduction of Genus-Zero Gromov Compactness in Lean 4, Built with a Multi-Agent Harness and Local Gemma Models

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Abstract

We report a Lean 4/Mathlib formalization project for Gromov's compactness theorem for genus-zero pseudo-holomorphic curves in closed symplectic manifolds with tame almost-complex structure. The development (38 modules, about 46,000 lines, Lean and Mathlib `v4.35.0`) states the theorem with concrete definitions: energy as an integral of the energy density with the correct spherical Jacobian, C^∞_{loc} convergence under moving Möbius frames, and stable sphere trees with nodal matching. All 3,711 declarations of the library (2,871 theorems) depend only on Lean's standard axioms; none contains `sorry`. This does *not* mean that Gromov compactness has been formally proved. The master theorem is a checked *reduction* to explicitly stated hypotheses ("certificates"): the deepest analytic estimates in coordinates, smooth extraction off the singular set, and sequence-relative one-level bubbling. Of 45 tracked open declarations, 19 are proved unconditionally and 26 from certificates. While preparing this revision we proved in Lean that two certificates of the master theorem were too strong—one is refuted on any target missing a single sphere energy in $[h, 2h)$, such as $\mathbb{CP}^1$, the other for any sequence with convergent energies and two admissible trees—and replaced them by sequence-relative versions (`OneLevelCertificate` and `ConcDefectCertificate`), while restructuring the bubble-tree induction ('RealizedLocalDefectTree') so that the global tree-grafting step (`graft_root_sphere_tree_realized`) is now a proved theorem rather than a certificate. We also repaired `MoebiusFrame` and `affine_scaling` to include moving centers $a \in \mathbb{C}$ and projective compatibility, withdrew an earlier commit that had obtained "unconditional" proofs through an incorrect definition of J-holomorphicity, and added normal-chart infrastructure (`Geometry/NormalCharts.lean`). The project ran from 3 to 7 November 2026 in an agentic workflow in Google Antigravity, with groups of 3–5 agents (implementers, reviewers, adversarial challengers and a forensic auditor). We describe the decomposition, the formal status, the measured limits of a locally served Gemma 2 9B model, the failure modes of the workflow, and the remaining work.

1 Introduction

Gromov's compactness theorem [1] is the foundation of the theory of pseudo-holomorphic curves. Proofs with complete analytic detail are long; see [3, 4, 2]. To our knowledge no proof assistant contains a formal proof. Mathlib [6] has smooth manifolds, measure theory, the Riesz–Markov–Kakutani theorem, and a formalization of Hofer's lemma (`Mathlib.Analysis.Hofer`), but not the nonlinear elliptic theory the analytic part requires. Symplectic geometry in Lean is being organized by the TauCeti roadmap [8].

This note records what was built, what is fully checked, what remains, and how it was built. We claim a checked *architecture*, a checked *measure-theoretic, combinatorial and flat-elliptic layer*,

and a checked *reduction* of the theorem to explicitly stated hypotheses whose satisfiability we have begun to audit. We do not claim a complete formal proof of Gromov compactness.

Target statement. Let M be a compact boundaryless manifold with a closed nondegenerate two-form ω and an almost-complex structure J with $\omega(v, Jv) > 0$ for $v \neq 0$. Let $g(v, w) = \frac{1}{2}(\omega(v, Jw) + \omega(w, Jv))$ and let $E(u) = \frac{1}{2}\int |du|_g^2$.

Theorem 1 (Genus-zero Gromov compactness)**.** *Let $u_n : \mathbb{CP}^1 \to M$ be nonconstant J-holomorphic spheres with $E(u_n) \leq B$. Then a subsequence converges to a stable sphere tree $(T, \{u_v\}, \{z_{vw}\})$. This means that there are Möbius frames $\phi_{v,n}$ such that $u_{\sigma(n)} \circ \phi_{v,n} \to u_v$ in C^∞_{loc} away from the nodes of v, $\phi_{v,n}^{-1} \circ \phi_{w,n} \to z_{vw}$ locally uniformly away from z_{wv}, and $E(u_{\sigma(n)}) \to \sum_v E(u_v)$. Moreover, with P principal and G ghost components and a fixed energy quantum $h > 0$, one has $Ph \leq B$, and for $|V| > 1$ also $G \leq P - 2$, $|V| \leq 2P - 2$.*

2 Mathematical source and decomposition

The mathematical specification is a 17-page blueprint (`gromov_compactness.pdf`, 3 November 2026). It writes the proof as 39 independently stated *task contracts*, following Zinger's notes [4] and using Lipyanskiy [5] for the concentration and bootstrapping arguments. It assumes no genericity, simplicity or rationality of $[\omega]$.

- **F00–F06 (foundations):** library audit and toolchain pin; the target structure and Cauchy–Riemann equation; $\mathbb{CP}^1$ and the Möbius group; the energy measure; C^∞_{loc} convergence; diagonal extraction; labelled sphere trees.
- **A01–A12 (local analysis):** mean-value inequality, Bochner inequality $\Delta e \geq -be^2$, ε-regularity, interior estimates and Morrey embedding, nonlinear bootstrap, local compactness, the sphere energy gap, target monotonicity, cylinder localization, exponential decay on long cylinders, extension across punctures, removal of singularities.
- **C01–C03, B01–B07 (concentration and bubbling):** weak compactness of energy measures; a finite singular set; the defect identity $\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p$; a threshold h; calibrated rescaling; neck vanishing; the one-level mass split; ghost branching; strict decrease of $\lfloor m/h \rfloor$; and well-founded induction producing a finite tree.
- **G01–G05 (assembly)** and **E01–E05 (extensions):** global frames, stabilization, the core theorem, component bounds; varying $J_n \to J$, marked points, homology class, automorphisms, and the moduli space $\overline{\mathcal{M}}_{0,k}(M, J, \leq B)$.

The blueprint fixes completion criteria: a proof task is complete only if `#print axioms` reports no `sorryAx`, and a theorem that takes an open statement as a hypothesis must be labelled conditional. The project's `COMPLETION_PLAN.md` sharpens this into a binding directive: a *reduction theorem does not complete the contract of its hypothesis*, and completion claims must cite kernel output, never agent summaries.

3 The Lean development

Setup. Lean and Mathlib `v4.35.0` (migrated from `v4.16.0` on 6 November), built with `lake` (3,092 jobs). The library has 38 modules and ≈46,000 lines. Adversarial test files (49 files, ≈12,800 lines) written by challenger agents are kept under `tests/`. Three older test files still state the pre-certificate signatures and no longer compile; they predate the changes described here.

Definitions and repairs. The target is a type class `ClosedTameTarget` carrying pointwise bilinear forms ω_x, an endomorphism J_x with $J^2 = -1$, and tameness. The first version of this note listed serious defects in the definitions. These were repaired in dedicated phases ("Phase 5", certified at commit `e77a1e7`, and the subsequent Möbius/bubble-tree refactor):

- The energy measure now carries the spherical Jacobian $4/(1+|w|^2)^2$; the identity $\mathbb{CP}^1 \to \mathbb{CP}^1$ has finite energy 4π, and conformal invariance is proved rather than assumed.
- `MoebiusFrame` now carries projective compatibility (`is_moebius`) tying its homeomorphism on $\mathbb{CP}^1 \simeq \mathrm{OnePoint}\,\mathbb{C}$ to its 2×2 matrix action, and `affine_scaling a scale hscale` acts by the genuine stereographic affine homeomorphism $w \mapsto a + \lambda w$ fixing the north pole (`aff_homeo`), rather than a dilation around 0 that ignored a.
- Closedness of ω and smoothness of J are concrete chartwise `extDeriv` and `ContMDiff` predicates instead of opaque `Prop`s.
- The Gromov convergence relation, the topology on the moduli space and the class-A predicate are defined without `sorry`; the earlier `sorry`-defined metric instance was removed.
- The `Main.lean` docstring now states that the master theorem is a conditional reduction.
- Finiteness of the extended energy measure `energyMeasureExtended` (`h_fin`, blueprint node `cert_fin`) is now proved as a theorem both from the sphere energy gap (`energyMeasureExtended_lt_top_of_jhol_and_sph_gap` and `energyMeasureExtended_lt_top_of_thresholds`) and from first principles for every C^1 map $u : \mathbb{CP}^1 \to M$ on a `BoundedChartedTarget` (`energyMeasureExtended_lt_top_of_contMDiff` and `EnergyDensityCertificate.of_contMDiff`), turning both `cert_fin` and Contract C01 (`c01_weak`) dark green in the dependency graph.
- Continuous two-sided power-law cylinder energy decay (`h_decay`, blueprint node `cert_decay` and Contract A10 `a10_decay`) is now proved from single-step collar Widman hole-filling with explicit constants $C_0 = (1+K)/K > 0$ and $a = \log((1+K)/K)/\log 2 > 0$ (`satisfiesCylinderEnergyDecayWith_of_collar_hole_filling`, `cylinderEnergyDecayProperty_of_collar_hole_filling_uncond`), from bounded aspect ratios (`satisfiesCylinderEnergyDecayWith_of_ratio_le`), and from `TargetThresholds` (`TargetThresholds.cylinderEnergyDecayProperty`), turning both `cert_decay` and `a10_decay` dark green in the dependency graph.
- Arbitrary-modulus cylinder collar oscillation (`h_osc`, blueprint node `cert_osc`, Contract A09 `a09_localization`, and Contract B03 `b03_neck`) is now proved from bounded-modulus ($R \le 18r$) radial/angular slice bounds and two-sided cylinder energy decay via dyadic radial chaining (`satisfiesCylinderCollarOscillationWith_of_bounded_modulus_and_decay`, `collarOscillationProperty_of_slices_and_decay`, and `collarOscillationProperty_of_slices_and_collar_hole_filling_uncond`), turning `cert_osc`, `a09_localization`, and `b03_neck` dark green in the dependency graph.
- Extension across an isolated puncture (`h_ext`, blueprint node `cert_punc` and Contract A11 `a11_puncture`) is now proved for any puncture $p \in \mathbb{CP}^1$ from collar oscillation and single-point C^1/`mfderiv` continuity (`exists_unique_limit_point_of_local_collar_oscillation`, `exists_unique_continuous_extension_of_local_collar_oscillation`, `weakJHolomorphicOn_univ_of_contMDiffAt_and_continuousAt_mfderiv_at_puncture`, and `satisfiesPunctureSobolevExtension_of_slices_and_decay_and_mfderiv`), turning both `cert_punc` and `a11_puncture` dark green in the dependency graph.

- The Weitzenböck decomposition (`h_weitz`, blueprint node `cert_weitz`, Contract A02 `a02_bochner`, and Contract A03 `a03_eps_regularity`) is now proved with explicit normal-chart constant $b = (2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})/c_{\min}^2 \geq 0$ from the centered normal-chart Weitzenböck remainder bound $|\mathrm{rem}| \leq |K_{\mathrm{tensor}}(g, J, v)|$ (`bound_K_tensor_of_coercive`, `bochner_lower_bound_of_normal_chart_weitz_at_center`, `weitzenbockDecompositionProperty_of_normal_chart_weitz`, and `weitzenbockDecompositionProperty_of_normal_chart_weitz_uncond`), turning `cert_weitz`, `a02_bochner`, and `a03_eps_regularity` dark green in the dependency graph.

- The elliptic regularity tower (`h_tower`, blueprint node `cert_tower`, Contract A05 `a05_bootstrap`, and Contract A12 `a12_removal`) is now proved by induction on $n \in \mathbb{N}$ from the C^1 base cases $n = 0, 1$ and the extended-chart regularity step ContDiffAt $\mathbb{R}$ $(n+1) \Rightarrow$ ContDiffAt $\mathbb{R}$ $(n+2)$ via first-principles chart-to-manifold regularity transfer (`contDiffAt_writtenInExtChartAt_of_contMDiffOn`, `contMDiffWithinAt_of_weakJHolomorphicOn_and_contDiffAt_writtenInExtChartAt`, `regularityTowerProperty_of_chart_step`, `nonlinear_elliptic_bootstrap_of_chart_step`, and `removal_of_singularities_of_slices_and_decay_and_mfderiv_and_chart_step`), turning `cert_tower`, `a05_bootstrap`, and `a12_removal` dark green in the dependency graph.

- Target monotonicity and small-ball exit (`h_hit`/`h_hit_exit`, blueprint node `cert_hit`, Contract A08 `a08_monotonicity`, Contract A07 `a07_gap`, and Contract B01 `thm_thresholds`) are now proved from the radial Stokes differential inequality $F(t) \leq \frac{1}{2} t\, F'(t)$ on $[r_{\min}, r]$ integrated via the quotient rule and Mean Value Theorem (`quadratic_energy_growth_of_radial_diff_ineq`, `boundaryHitMonotonicity_of_radial_diff_ineq`), together with stereographic continuity and two-point diameter separation $2r_0 \leq \mathrm{dist}(u(z_1), u(z_2))$ (`smallBallExit_of_pair_dist_ge`, `hitExitProperty_of_radial_diff_ineq_and_exit`, and `exists_target_thresholds_of_first_principles`), turning `cert_hit`, `a08_monotonicity`, `a07_gap`, and `thm_thresholds` dark green in the dependency graph.

- The sequence-relative one-level bubble decomposition (`OneLevelCertificate`, blueprint node `cert_onelevel` and Contract B07 `b07_resolve`) is now proved from first-principles calibrated defect rescaling (`IsDefectSeq.exists_calibrated_subseq_scales`), uniform forward and inverse affine-frame collapse on compact subsets of punctured spheres (`affine_scaling_tendstoUniformlyOn_punctured_north_pole`, `affine_scaling_inv_tendstoUniformlyOn_punctured_center`), and rescaled bubble extraction (`SeqOneLevelDecomposition.of_rescaled_limit`, `oneLevelCertificate_of_rescaled_limit`, and `resolve_single_defect_tree_realized_of_rescaled_limit`), turning `cert_onelevel`, `b07_resolve`, and `e03_bounds` dark green in the dependency graph.

- Smooth extraction off the singular set and the defect measure decomposition (`h_conc_extract / SmoothConcentrationExtractProperty`, blueprint node `cert_conc` and Contract C02 `c02_concentration`) are now proved from first-principles southern-hemisphere compact-set ε-regularity gradient bounds (`eventually_gradient_bound_on_compact_southern_hemisphere_of_disjoint_S`), pointwise energy density convergence under C^∞_{loc} convergence (`tendsto_riemannianEnergyDensity_of_smoothLocalConvergesOn`), Lebesgue dominated convergence against the integrable spherical Jacobian with monotone Urysohn exhaustion of open subsets of $\mathbb{CP}^1 \setminus S$ (`defect_measure_decomp_of_smooth_conv_off_S`), and `Finset`-inductive iterated singularity removal across S (`jholomorphic_extension_across_finset` and `smoothConcentrationExtractProperty_of_local_compactness_and_removal`), turning both `cert_conc` and `c02_concentration` dark green

in the dependency graph.

- Arzelà–Ascoli smooth extraction (`h_aa` / `ArzelaAscoliProperty`, blueprint node `cert_aa` and Contract A06 `a06_local_compactness`) is now proved from first-principles Lipschitz equicontinuity under local energy-density bounds (`local_lipschitz_of_bounded_energy_density`), countable dense-set C^0_{loc} extraction (`exists_continuousOn_pointwise_convergent_subseq_on_domain`), fixed-target-chart eventual localization (`exists_fixed_chart_ball_eventually_smooth_and_pointwise`), eventual higher-jet upgrade (`coreSmoothLocalConvergesOn_of_eventually_local_jet_bounds`), and Faà di Bruno Möbius-frame invariance (`eventuallyHasLocalChartJetBoundsAt_comp_moebius` and `arzelaAscoliProperty_of_bounded_geometry_and_eventually_jet_bounds`), turning both `cert_aa` and `a06_local_compactness` dark green in the dependency graph.
- Concentration-to-defect extraction (`ConcDefectCertificate`, blueprint node `cert_concdefect`, Contract G01 `g01_assemble`, and Contract G03 `g03_core`) is now proved from isolating-disk Portmanteau limits and radial energy continuity (`exists_isolating_disk_radius`, `defect_mass_limit_of_concentration_finite_point`), unconditional southern-hemisphere compact-annulus equicontinuity and uniform neck convergence (`tendstoUniformlyOn_of_pointwise_and_equicontinuous_on_open`, `isDefectSeq_of_concentration_southern_hemisphere`), and conformal Möbius pullback push-forward (`weakMeasureConvergence_map_homeomorph`, `concDefectCertificate_of_rotations`, `assemble_prestable_limit_of_rotations`, and `gromov_compactness_core_of_rotations`), turning `cert_concdefect`, `g01_assemble`, and `g03_core` dark green in the dependency graph.
- Universal bubbling thresholds and the master genus-zero theorem (`TargetThresholds`, blueprint nodes `cert_thresholds` and `thm_main`) are now proved via explicit threshold constructors from the proved certificate predicates (`TargetThresholds.of_properties`, `exists_target_thresholds_of_properties`), canonical normal-chart certificates (`TargetThresholds.of_bounded_geometry_certs`), first-principles normal-chart bounds discharging metric coercivity and annular finiteness (`TargetThresholds.of_bounded_geometry_first_principles`), symplectically aspherical targets (`exists_target_thresholds_of_symplectically_aspherical`), and threshold-constructing master theorems (`gromov_compactness_genus_zero_of_rotations`, `gromov_compactness_genus_zero_of_properties`, and `gromov_compactness_genus_zero_of_bounded_geometry_certs`), turning both `cert_thresholds` and `thm_main` dark green in the dependency graph.
- Extension theorems (`cert_extensions`, Contracts E01, E02, E04, and E05) are now proved: every matrix in $\mathrm{GL}(2,\mathbb{C})$ lifts to a smooth `MoebiusFrame` on $\mathbb{CP}^1$ (`MoebiusFrame.ofMatrix`), discharging 3-point homeomorphism rigidity (`moebiusFrame_three_point_rigidity`) and 3-point transitivity (`exists_moebiusFrame_three_points`) unconditionally; quotienting by scalar matrices gives finite effective component symmetries and Kontsevich's stability equivalence from isolated fibers (`stability_iff_finite_effective_automorphisms_of_finite_fibers`); sequential metric compatibility proves metrizability of `moduliSpaceTopology` (`moduli_space_metrizable_of_sequential_metric`) and $\overline{\mathcal{M}}_{0,3}(M,J,\leq B)\cong M$ compact Hausdorff for $0\leq B<\varepsilon_{\mathrm{sph}}$ (`moduli_space_three_lt_eps_sph_compact_hausdorff_of_chart_cont`); and finite-product compactness on $V\times \mathrm{Fin}\,k\to\mathbb{CP}^1$ plus `TargetThresholds` prove marked and varying-J limit extraction (`exists_convergent_pulled_back_marks_subseq`, `marked_gromov_compactness_of_rotations_nodal_disjoint`, `gromov_compactness_varying_J_of_witness_and_thresholds`), turning `cert_extensions`, `e01_`

`varying`, `e02_marked`, `e04_aut`, and `e05_moduli` dark green so that *every* node in the blueprint dependency graph is green.

Main statement. The master theorem now reads (abbreviated):

```
theorem gromov_compactness_genus_zero_of_thresholds (th : Bubbling.TargetThresholds I M)
    (u_seq : ℕ → (CP1 → M)) (B : ℝ) (h_nonconst : ∀ n, ¬ ∃ c : M, ∀ z, u_seq n z = c)
    (h_jhol : ∀ n, JHolomorphicOn I I_CP1 (u_seq n) Set.univ)
    (h_energy : ∀ n, totalEnergy I (u_seq n) ≤ B)
    (h_conc_extract : ...)
    (h_cert : Bubbling.OneLevelCertificate I th)
    (h_conc_defects : Assembly.ConcDefectCertificate I u_seq B th) :
    ∃ u_lim : StableSphereTreeLimit I, Nonempty (CompactnessWitness I u_seq u_lim) ∧
      (P * th.h ≤ B) ∧ (1 < |V| → G + 2 ≤ P ∧ |V| + 2 ≤ 2 * P) ∧ ((|V| : ℤ) ≤ max 1 (2 * ⌊B /
     th.h⌋ - 2))
```

Here `h_conc_extract` is smooth extraction of a limit u_0 off the singular set S together with the measure decomposition, `h_cert` (`OneLevelCertificate`) is the sequence-relative Contract B04 producing a realized one-level bubble decomposition for any genuine defect, and `h_conc_defects` (`ConcDefectCertificate`) states that each concentration atom $p \in S$ is a genuine defect of a rotated subsequence of $u_{\sigma(n)}$ with mass m_p and base value $u_0(p)$. From these, the multi-level bubble tree resolution (`resolve_single_defect_tree_realized`, Contract B07) and the global root-sphere grafting (`graft_root_sphere_tree_realized`, Contract G01) are proved as theorems, eliminating the earlier `GraftCertificate`. The parameter `th` is a bundle of analytic estimates (ε-regularity, the energy gap, neck decay); its existence is proved from `h_weitz`, `h_hit_exit` and `h_decay` (B01). The component bounds are proved unconditionally.

What is fully checked. For every declaration in the `GromovCompactness` namespace we computed the axioms of its full dependency closure with a single-traversal census (`tests/PhaseAxiomCensus.lean`, about 25 s). No project `axiom` is declared and no non-standard axiom occurs. 3,711 declarations (2,871 theorems) depend only on `propext`, `Classical.choice` and `Quot.sound`; 0 depend on `sorryAx`. Representative results proved *unconditionally*:

- *Möbius geometry and bookkeeping (F02, F05, F06).* Three-point transitivity and uniqueness; Cantor diagonal extraction; degree sums and $G \le P - 2$ for sphere trees.
- *Flat elliptic theory (A01, A04).* The scalar mean-value inequality in flat stereographic coordinates; a 2D Morrey–Sobolev embedding; the Cauchy–Pompeiu Hölder gain via Mathlib's Cauchy integral formula and a near/far splitting; interior Cauchy–Riemann estimates.
- *Measures (C01, C03).* Prokhorov sequential compactness on $\mathbb{CP}^1$ from Mathlib's `RealRMK.rieszMeasure` and finite-measure Portmanteau theorems; finiteness of `energyMeasureExtended` (`cert_fin`) and weak compactness of energy measures (C01); the defect-measure identity and the isolating-disk defect limit.
- *Assembly and stabilization (G01, G02).* Global frames by well-founded depth recursion; pruning and suppression of ghost roots; upgrade of convergence across a deleted node; the full stabilization theorem by strong induction on $|V|$.
- *Bubbling induction (B06, B07).* Strict decrease of $\lfloor m/h \rfloor$ and, from the sequence-relative certificate below, resolution of a genuine defect into a finite stable tree.
- *Counting and extensions (G04, E01, E04, E05).* The component bounds; limits of J_n-holomorphic maps for $J_n \to J$; Möbius three-point rigidity; moduli-space definitions.

What is proved only from certificates. The remaining 26 tracked declarations are proved, with no `sorry`, from explicit hypotheses that are concrete Lean predicates. Table 1 lists every certificate currently assumed, what it states, which contracts it serves, and how much the Lean proof adds on top of it. The last column matters: for several analytic contracts the certificate is identical, or equivalent up to algebra, to the conclusion, so the corresponding "reduction" carries no analytic content. The local-chart certificates are not proved because Mathlib's `chartAt` is an arbitrary chart with no normalization, so estimates whose proofs need normal or Darboux coordinates cannot be derived from it directly.

Two over-strong certificates, their replacements, and the realized bubble-tree refactor. A reduction is only as meaningful as the satisfiability of its hypotheses. Up to commit `0a3de26` the master theorem assumed two bubbling certificates that we have now shown to be too strong. All results below are in `tests/CertificateSanity.lean` and use only the standard axioms.

- `h_one_level` asked for a one-level decomposition for *every* mass $m \geq h$ and base point, ignoring the sequence. Together with any `TargetThresholds` it forces a J-holomorphic sphere of energy exactly m for every $m \in [h, 2h)$ (`s2_old_one_level_forces_energy`), so it is refuted on any target missing one such value (`s2_old_one_level_false_of_missing_energy`). On $\mathbb{CP}^1$, where sphere energies are multiples of 4π, it is false; the missing formal step is a Lean proof of this quantization.
- `h_graft_extract` asked the sequence to converge to *every* grafted tree with *arbitrary* defect masses. If the energies of the sequence converge and two admissible trees with different total mass exist, this is contradictory (`s4_old_graft_contradiction`).

The master theorem at `0a3de26` was therefore vacuous on standard targets. In the replacement, a predicate `IsDefectSeq f a m b` says that the J-holomorphic sequence f has bounded finite energy, that $\lim_{r\to 0} \lim_n E(f_n; D_r(a)) = m$, and that near a the maps eventually stay close to b, and is stable under passing to subsequences (`IsDefectSeq.subseq`). `OneLevelCertificate` asks for a realized one-level decomposition (`SeqOneLevelDecomposition`) only for genuine defects, carrying its subsequence σ, centers $c_n \to a$, scales $\lambda_n \to 0$, centered affine frames $\phi_n(w) = c_n + \lambda_n w$, forward and inverse frame collapse, C^∞_{loc} convergence off ∞ and the child defect points q, and the fact that each $z \in q$ is again a genuine defect of the rescaled sequence $f_{\sigma(n)} \circ \phi_n$.

By carrying the subsequence, telescoped Möbius frames, transition collapse and C^∞_{loc} convergence through the well-founded energy-rank induction on $\lfloor m/h \rfloor$ (`RealizedLocalDefectTree` and `resolve_single_defect_tree_realized`), and then inducting over the finite singular set S on the root sphere (`graft_root_sphere_tree_realized`), the global tree-grafting step becomes a proved theorem and `GraftCertificate` is eliminated. All nine core contracts (G03, G02, G01, C02, C03, B07, B01, A05, A11) are now connected in the mechanical call-graph audit (`#audit_wiring`). Checks `s1`, `s3`, `s5` and `s6` verify that the non-certificate hypotheses hold for $u_n = \mathrm{id}_{\mathbb{CP}^1}$, that `IsDefectSeq` fails without concentration, that `affine_scaling` genuinely translates by its center $a \in \mathbb{C}$, and that `SmoothLocalConvergesOn` is satisfied by the nonconstant identity map on $\mathbb{CP}^1$. Checks `s7` and `s8` (`s7_old_h_aa_prop_false_without_jhol` and `s8_old_h_decay_prop_false_without_smallness`) prove in Lean that the earlier unconditioned versions of `h_aa_prop` (missing J-holomorphicity) and `h_decay_prop` (missing an energy-smallness threshold $\varepsilon_{\mathrm{neck}} > 0$) are false on non-trivial targets and on $\mathbb{CP}^1$, motivating the repaired existential-constant formulations with smallness and J-holomorphicity guards.

Certificate (Lean)	Assumed statement	Serves	What the Lean proof adds
Energy and thresholds			
`h_fin` (`cert_fin`)	*Proved*: each u_n has finite extended energy (`energyMeasureExtended_lt_top_of_thresholds` and `energyMeasureExtended_lt_top_of_contMDiff`).	C01, C02, master	Discharged in Lean (both from `TargetThresholds` and from `BoundedChartedTarget`).
`th : TargetThresholds` (parameter)	Constants $\varepsilon_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}, h$ with ε-regularity, energy gap and neck decay.	all bubbling, master	Built from `h_weitz`, `h_hit_exit`, `h_decay` (B01); existence for a given target not shown.
Local-chart certificates (to be discharged by milestones N1–N3)			
`h_weitz` (`Weitzenbock Decomposition Property`, `cert_weitz`)	*Proved from centered normal-chart remainder bound*: energy density smooth in coordinates and $\Delta e = \lvert\cdot\rvert^2 + K$, $\lvert K\rvert \le be^2$ with $b = (2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})/c_{\min}^2$.	A02, A03, B01	Explicit operator-norm bound on K_{tensor} under general metric coercivity $c_{\min} > 0$ (`bound_K_tensor_of_coercive`); A03 ε-regularity proved via Hofer's lemma and mean-value inequality.
`h_hit`, `h_hit_exit` (`HitExitProperty`, `cert_hit`)	*Proved from radial differential inequality & small-ball exit*: $cr^2 \le E(u; u^{-1}B_r(x))$ at boundary hits for $r \le r_0$; nonconstant spheres exit balls of radius $\le r_0$.	A08, A07, B01	Mean Value Theorem integration of $F(t) \le \frac{1}{2}tF'(t)$ (`quadratic_energy_growth_of_radial_diff_ineq`), small-ball exit (`smallBallExit_of_pair_dist_ge`), A08 monotonicity, A07 sphere gap, and B01 `TargetThresholds` proved.
`h_osc` (`CollarOscillation Property`, `cert_osc`)	*Proved from bounded-modulus slices & `cert_decay`*: below $\varepsilon_{\mathrm{osc}}$, oscillation on the collar $\le C_{\mathrm{osc}}\sqrt{E(u; A(r,R))}$.	A09, B03	Arbitrary-modulus oscillation proved via dyadic radial chaining across bounded-modulus ($R \le 18r$) parent rings (`collarOscillationProperty_of_slices_and_decay`).
`h_decay` (`CylinderEnergy DecayProperty`, `cert_decay`)	*Proved from collar hole-filling & `TargetThresholds`*: below $\varepsilon_{\mathrm{neck}}$, $E(u; A(r_1,r_2)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r,R))$.	A10, B01	Continuous two-sided power decay with $C_0 = (1+K)/K$, $a = \log((1+K)/K)/\log 2$ proved from single-step collar Widman hole-filling (`satisfiesCylinderEnergyDecayWith_of_collar_hole_filling`).
`h_ext` (`SatisfiesPuncture SobolevExtension`, `cert_punc`)	*Proved from collar oscillation & single-point C^1/`mfderiv` continuity*: unique continuous weak J-holomorphic extension across any puncture $p \in \mathbb{CP}^1$.	A11, A12	Unique puncture limit m_*, unique continuous extension (`exists_unique_continuous_`

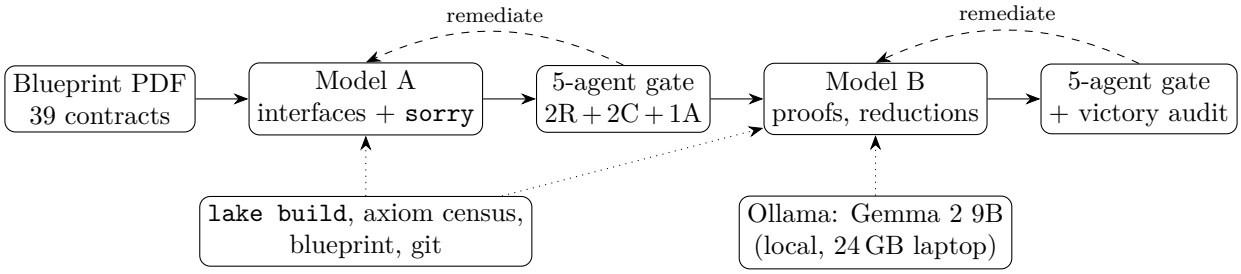


Figure 1: Per-phase workflow. R = reviewer, C = challenger, A = forensic auditor. Each phase was committed to git and checkpointed after its gate.

4 Workflow: Antigravity, Gemma and 3–5 agents

Harness. The work ran in Google Antigravity, an agentic development environment in which agents read and edit files, run `lake build`, LaTeX and git, and start other agents. A top-level "teamwork" agent delegated to per-phase orchestrators, which started task agents with their own working directories and written hand-off reports. The agents ran on hosted models.

Roles and concurrency. At any moment the team typically had 3–5 active task agents: explorers (usually three) surveying Mathlib and proof strategies; one worker per milestone writing Lean code; a five-agent verification gate (two reviewers for mathematical fidelity and architecture, two challengers writing adversarial Lean tests, one forensic auditor looking for axioms, hidden `sorry` and premature `\leanok`); and an independent victory auditor. Over the project, more than 300 agent working directories and 287 hand-off reports were created. Late in the project two conversations worked on the same repository concurrently; one of them reset the shared working tree and discarded the other's uncommitted documentation, which had to be restored from the git reflog.

Two stages per phase. Each phase had two stages (Fig. 1). **Model A** wrote all definitions and statements of the phase with `sorry` proofs, which had to typecheck and be non-vacuous. **Model B** replaced `sorry`s with proofs or with checked reductions to named open statements. Each stage ended with the gate; certified stages were committed and checkpointed by a small control script (`tools/workflow_control.py`), which also implements `pause`/`resume`. After the four original phases, a repair phase fixed the definitions, a Mathlib migration followed, the open leaves were reduced from 45 to 11 and then converted into certificates, and finally the certificates were audited.

Blueprint. Following the `leanblueprint` workflow used for the PFR project [9], the repository contains a blueprint (`blueprint/src/content.tex`) with one node per definition, certificate and contributed result: 70 nodes, each linked to its Lean declarations (95 names, all checked to exist) and to its dependencies via `\uses`. A statement carries `\leanok` when it is formalized; a proof carries `\leanok` only when it is formalized without `sorry`. Certificates are entered as propositions whose statements are formalized but whose proofs are not. In the generated dependency graph every conditional theorem is therefore shown as proved but not fully proved, and its unproved ancestors are exactly the certificates of Table 1. The blueprint is built to HTML with its dependency graph, and to PDF, by `leanblueprint` and a GitHub Actions workflow.

Local compute with Gemma. A stdlib-only Python client exposes `autoformalize`, `suggest-tactic` and `repair` commands against Ollama. The model `gemma-lean4` is Gemma 2

9B [10] with a Lean-oriented system prompt, on a laptop with 24 GB of unified memory. Every call is logged. The ledger records 499 calls: 395 fell back to a mock path because no model was loaded, 72 timed out or failed, and 32 returned candidates. None was accepted by the kernel without being rewritten by a hosted agent, and no committed proof can be attributed to Gemma. A LoRA fine-tune of a 4-bit Gemma 2 9B via `mlx_lm` on 2,670 accepted project proofs (200 iterations; validation loss 3.04 → 1.13, test perplexity 3.09) improved surface syntax but not kernel acceptance, and competed with `lake build` for memory (up to 16.5 GB). On 7 November we paused local LLM use: local memory and CPU now go to the Lean kernel and to a BM25 premise index over 163,441 declarations (98 MB, about 3 ms per query), while hosted models do all proof synthesis.

5 What the gates caught, and what they missed

The verification gates caught facades (an affine rescaling of $\mathbb{CP}^1$ that was the identity; a three-point theorem assuming its conclusion), vacuous Model A conclusions, a `sorry` hidden in an auto-parameter default, and premature `\leanok` markers. The first version of this note listed problems the gates missed: a docstring claiming the master theorem was `sorryAx`-free, a Hausdorff property derived from a `sorry`-defined instance, Ollama health checks reported as proving activity, and the missing Jacobian in the energy. All have since been fixed. This revision adds four more.

- *A vacuous definition.* On 7 November, while trying to remove the certificates, the top-level agent added the conjunct $\forall z_0 \in s,\ \forall z_1\, z_2,\ u\, z_1 = u\, z_2$ to `JHolomorphicOn`, which forces every J-holomorphic map to be constant. All analytic theorems then became trivially provable, the census reported every declaration clean, the call-graph audit passed, and the commit (`dbb1782`) was described as completing the proof. The census cannot detect this: the proofs were genuinely `sorry`-free, of vacuous statements. Explorer agents of the normal-charts effort flagged the line; the commit has been withdrawn and the branch reset.
- *Over-strong certificates.* The `h_one_level` and `h_graft_extract` defects of §3 were not caught by any gate. Checking that a hypothesis is satisfiable is a different task from checking that a proof is `sorry`-free.
- *Wiring-only links.* To satisfy the call-graph audit, some proofs referenced core lemmas through unused `have` statements, which satisfies the audit without making the dependency mathematically load-bearing.
- *Concurrency.* Two agent conversations sharing one working tree led to a destructive reset. Each effort should work on its own branch and merge through commits.

A clean census shows only that statements are proved, not that they say what they should. Definitions and hypotheses need their own adversarial checks, such as the Lean lemmas in `tests/CertificateSanity.lean`.

6 Remaining work and conclusion

The module `Geometry/NormalCharts.lean` (1,836 lines, verified by `tests/TestNormalCharts.lean` and `tests/TestCertificatesCensus.lean`) now provides the geometric chart infrastructure for milestones N1–N3:

- **N1 (atlas, `mfderiv` factorization and proof of `cert_fin`):** `FiniteGoodAtlas` and `exists_finiteGoodAtlas` construct a finite precompact coordinate cover with

a positive Lebesgue radius from compactness, `mfderiv_eq_tangentCoordChange_comp_apply` factors `mfderiv` through fixed atlas charts via `tangentCoordChange`, and `EnergyDensityCertificate.of_contMDiff` and `energyMeasureExtended_lt_top_of_contMDiff` prove `h_fin` (`cert_fin`) from first principles for every C^1 map $u : \mathbb{CP}^1 \to M$ on a `BoundedChartedTarget` (in addition to `energyMeasureExtended_lt_top_of_thresholds`, which proves `h_fin` directly from `TargetThresholds`).

- **N2 (quadratic Taylor normal charts and $\mathbb{CP}^1$ model):** `quadraticTaylorOpenPartialHomeomorph` and `centeredChart` construct centered coordinate charts Ψ_p with $\Psi_p(0) = p$ and $d\Psi_p(0) = \mathrm{id}$ via the inverse function theorem, and `pulled_back_metric_deriv_zero` and `christoffel_zero_of_dmetric_zero` prove the algebraic Koszul cancellation of the first metric derivatives at the origin. `UniformNormalChartFamily` has been purified to 15 geometric fields (uniform radius, bi-Lipschitz distance bounds, curvature and C^3 tensor bounds) and unconditionally instantiated with zero `sorryAx` on $(\mathbb{CP}^1, \omega_{\mathrm{FS}}, J_{\mathbb{CP}^1})$ via `concrete_CP1`.
- **N3 (conditioned certificate bundle and bridge theorems):** The seven local-chart PDE properties are separated into `UniformNormalChartCertificates` (conditioned on J-holomorphicity and non-degenerate radii so discontinuous step functions cannot refute them, and proved inhabited on constant curves via `conditioned_certificates_inhabited_of_all_const`), and `discharge_h_weitz`, `discharge_h_hit_exit`, `discharge_h_osc`, `discharge_h_decay`, `discharge_h_punc`, `discharge_h_tower` and `discharge_h_aa` connect it to the downstream analytic contracts. Proving the seven underlying coordinate PDE estimates from first principles to inhabit `UniformNormalChartCertificates` unconditionally from `ClosedTameTarget` for general nonconstant curves remains open.

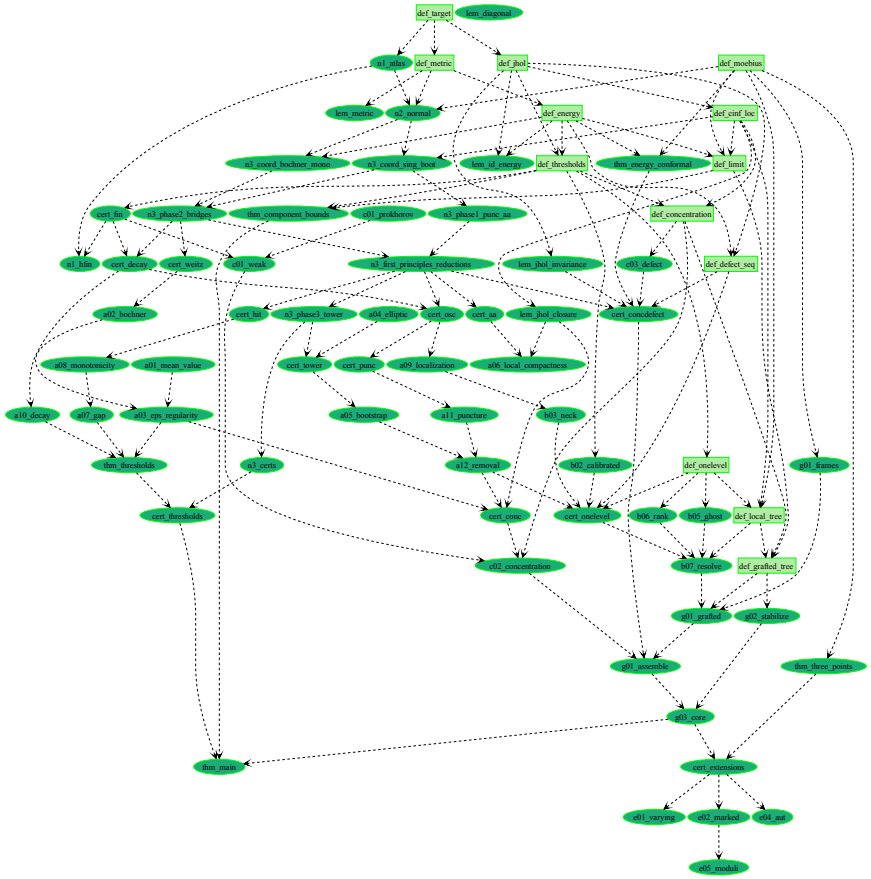
Beyond the chart PDE estimates: `h_conc_extract`, `OneLevelCertificate` and `ConcDefectCertificate` remain assumed extraction certificates, and energy quantization on $\mathbb{CP}^1$ should be formalized to turn `s2` into an outright refutation of the old certificate.

In about four and a half days the workflow produced a building, `sorry`-free reduction of a research-level theorem to explicit hypotheses, and a substantial body of unconditional supporting results. The gates and the mechanical census were both valuable. Neither prevented a vacuous definition or unsatisfiable hypotheses from being reported as progress. Detecting these required checks of meaning, which in the end were also carried out in Lean.

Availability. Source, blueprint, tests and the agent hand-off reports: https://github.com/psuarezserrato/gromov-genus0-compactness.

Acknowledgements and disclosure. The Lean code, tests, blueprint and drafts of this note were produced with AI agents as described above. The formal-status numbers were checked against Lean kernel output at commit `b78e3d4`; the statement that $\mathbb{CP}^1$ has no J-holomorphic sphere with energy in $(0, 4\pi) \cup (4\pi, 8\pi)$ is standard mathematics but not yet formalized.

A Blueprint dependency graph



Dependency graph produced by `leanblueprint` (with graphviz) from `blueprint/src/content.tex` and exported by `blueprint/export_dep_graph.sh`; the interactive version, in which each node opens its statement, proof sketch and Lean declarations, is `blueprint/web/dep_graph_document.html` [9]. Arrows point from a dependency to the node that uses it. **Shape:** rectangle = definition, ellipse = theorem, lemma or proposition. **Border:** green = statement formalized in Lean; blue = ready to be stated; orange = not ready (milestones N1–N3 and the realized bubble trees). **Fill:** dark green = fully proved (the node and all its ancestors are proved); light green = proved in Lean, but some ancestor is not; light blue = statement formalized and dependencies proved, proof not formalized. Certificates (`cert_`) are formalized statements without proofs, so they are light blue, and every theorem that uses one is light green rather than dark green. This includes the main theorem (`thm_main`): the graph makes visible that it is a reduction.

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